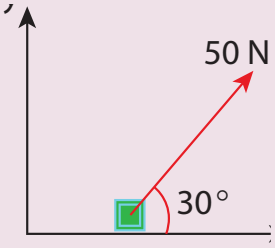
www.tntextbooks.in

IV. Conceptual Questions

1. Why it is not possible to push a car from inside?
2. There is a limit beyond which the polishing of a surface increases frictional resistance rather than decreasing it why?
3. Can a single isolated force exist in nature? Explain your answer.
4. Why does a parachute descend slowly?
5. When walking on ice one should take short steps. Why?
6. When a person walks on a surface, the frictional force exerted by the surface on the person is opposite to the direction of motion. True or false?
7. Can the coefficient of friction be more than one?
8. Can we predict the direction of motion of a body from the direction of force on it?
9. The momentum of a system of particles is always conserved. True or false?

V. Numerical Problems

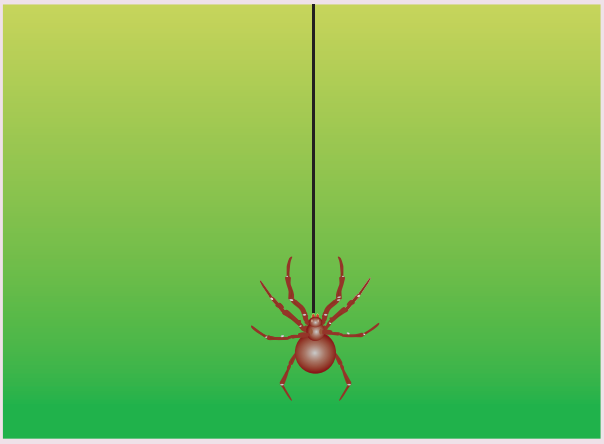
1. A force of 50N act on the object of mass 20 kg. shown in the figure. Calculate the acceleration of the object in x and y directions.



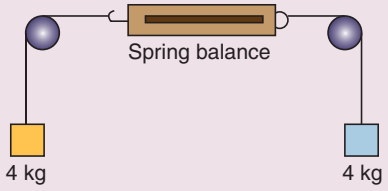
Ans: a_x=2.165 ms^{-2}; a_y=1.25 ms^{-2}

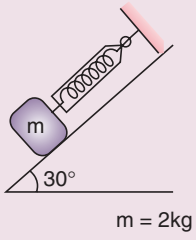
2. A spider of mass 50 g is hanging on a string of a cob web as shown in the figure. What is the tension in the string?

Ans: T=0.49N



3. What is the reading shown in spring balance?





Ans: Zero, 9.8 N

4. The physics books are stacked on each other in the sequence: +1 volumes 1 and 2; +2 volumes 1 and 2 on a table.
   a) Identify the forces acting on each book and draw the free body diagram.
   b) Identify the forces exerted by each book on the other.
5. A bob attached to the string oscillates back and forth. Resolve the forces acting on the bob in to components. What is the acceleration experience by the bob at an angle θ.